Helium - Technical Specification v3.1

Module 1: Executive Overview & Phased Roadmap

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Document Version History

Version	Date	Changes
1.0	Oct 2024	Initial specification (pre-mandate)
2.0	Oct 2025	Enhanced workflows, Portobello, extended document types
3.0	Nov 2025	Updated post-November 2025 mandate; reflected current compliance reality; revised timelines and positioning
3.1	Nov 2025	Corrected IRN/QR generation flow; clarified dual AR/AP capabilities; enhanced acceptance workflow; added consultation services; refined Enterprise tier features

Helium Editions & Version Mapping

Helium is available in three editions, each aligned with platform versions:

EDITION	VERSION	FEATURES
Helium Standard	v1.0 Only	Core compliance capabilities Invoice submission & IRN Basic notifications ERP write-back
Helium Professional	v1.0 + v1.1	All Standard features, PLUS: Automated acceptance workflow Dual-channel notifications Portobello buyer onboarding CRM integration
Helium Enterprise	v1.0 → v2.0 (Complete)	All Professional features, PLUS: Dedicated Taxly gateway (priority processing with reduced queue times) AI-powered data mapping Predictive analytics Self-service buyer portal Advanced reporting Optional VPN direct connection to Taxly (starting from ₦4M/year) SLA monitoring included

This specification documents Helium Enterprise capabilities (complete platform). Enterprise tier includes dedicated Taxly API gateway ensuring priority processing with dedicated channel for reduced queue times. Edition comparison and pricing details are provided in Module 10.

Important Notes

Post-Mandate Update (November 2025):
This specification has been updated to reflect the current regulatory environment following the November 2025 FIRS e-invoicing mandate. The mandate is now in effect. Language, timelines, and positioning throughout this document acknowledge the current compliance requirements.

Preview Document Scope:
This document contains Module 1 (Executive Overview & Phased Roadmap) only. Two additional documents are provided separately for evaluation:

- **Module 2: Security & Compliance Architecture** (separate document)
- **Complete Technical Specification: Modules 1-11** (provided upon formal engagement with WestMetro Limited)

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Module 1: Executive Overview and Phased Roadmap

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1.1 What Is Helium?

Enterprise-Grade E-Invoice Orchestration Platform

Helium is a middleware platform that deploys within your infrastructure as an extension of your existing environment. Operating independently from your ERP's core systems, Helium orchestrates the complete invoice compliance lifecycle—handling both Accounts Receivable workflows when you issue invoices to buyers, and Accounts Payable workflows when you receive invoices from suppliers.

Helium automatically transforms invoice data into FIRS-compliant UBL 3.0 JSON, generates Invoice Reference Numbers (IRNs) and QR codes, orchestrates all regulatory interactions with FIRS through the Taxly Access Point Provider (APP), manages buyer acceptance workflows, and synchronizes results back to your ERP—all while your business systems remain untouched. Whether you operate SAP, Dynamics 365, Oracle, Quantum, or custom-built systems, Helium provides a universal compliance layer under your full operational control through APIs or file-based transfers.

Dual Workflow Orchestration:

- **Accounts Receivable (Seller Side)**: When your ERP generates invoices, Helium creates IRNs, submits to FIRS for validation, and manages the buyer acceptance lifecycle
- **Accounts Payable (Buyer Side)**: When suppliers invoice you, Helium receives webhook notifications

from Taxly, initiates acceptance workflows, and orchestrates your response within the 72-hour acceptance window

Helium is SMTP-aware, enabling bidirectional email communication for notifications, acceptance requests, and reminders—ensuring seamless interaction with trading partners regardless of their technical infrastructure.

Key Value Propositions

- **Zero ERP Modification**: Your core business systems remain untouched—Helium operates independently from your ERP architecture, preserving system stability and vendor support agreements.
- **Universal Compatibility**: Works with any ERP system through flexible integration patterns (API or SFTP), eliminating vendor lock-in and supporting future ERP migrations.
- **Complete Lifecycle Orchestration**: Helium manages both outbound invoices (AR workflow) and inbound invoices (AP workflow) in a single platform, providing unified compliance management.
- **Zero-Downtime Architecture**: Queue-based message persistence ensures invoices are never lost during system maintenance or infrastructure events. Enterprise editions support optional high-availability configurations for continuous 24/7 operations.
- **Compliance Automation**: Handles IRN generation, transformation, submission, acceptance tracking, reminders, and notifications automatically—reducing manual processes and compliance risks.
- **Phased Delivery**: Begin with core compliance implementation (v1.0, available 8-12 weeks from contract signing), gain workflow automation in Q1 2026 (v1.1), and unlock AI-powered intelligence in Q2 2026 (v2.0)—all included in Helium Enterprise subscriptions.

System Architecture Overview

```
┌──────────────────────────────────────────────────────────────────────────┐
│                    YOUR INFRASTRUCTURE (Your Control)                    │
│                                                                          │
│  ┌──────────────────┐         ┌──────────────────────────┐              │
│  │                  │         │                          │              │
│  │    Your ERP      │────────►│         HELIUM           │              │
│  │                  │ Invoice │                          │              │
│  │  - Core Logic    │  Data   │  - IRN/QR Generation     │              │
│  │  - Database      │         │  - Data Transformation   │              │
│  │  - Business Rules│         │  - Workflow Orchestration│              │
│  │    (Untouched)   │◄────────│  - Acceptance Management │              │
│  │                  │         │  - Notification Engine   │              │
│  └──────────────────┘ Status  │  - SMTP Email Handler    │              │
│                       Updates └──────────────────────────┘              │
│                                       │            ▲                     │
└───────────────────────────────────────┼────────────┼─────────────────────┘
                                        │            │
                                        ▼            │
                              ┌──────────────────────────┐
                              │    Taxly (APP) / FIRS    │
                              │  - Invoice Validation    │
                              │  - Digital Signing       │
                              │  - Webhook Events        │
                              │  - Status Polling (Fallback) │
                              └──────────────────────────┘

                                        │
                                        │  Invoice Transmission
                                        ▼
                              ┌──────────────────────────┐
                              │      Buyer's APP → Buyer │
                              │  - Receive notifications │
                              │  - Accept/Reject         │
                              └──────────────────────────┘
                                        │
                                        | Acceptance Response
                                        ▼
                        (Back to Taxly → Helium via webhooks)

NUMBERED FLOWS (Accounts Receivable - Seller Workflow):
① ERP exports invoice data to Helium (API/SFTP)
② Helium generates IRN and QR code
③ Helium submits invoice to Taxly/FIRS for validation
④ FIRS returns validation status and digital signature
⑤ Taxly transmits invoice to buyer's APP (buyer receives notification from their APP)
⑥ Buyer accepts/rejects via their APP or FIRS portal
⑦ Buyer's APP/FIRS sends acceptance status to Taxly
⑧ Taxly webhooks notify Helium of acceptance status
⑨ Helium writes back acceptance status to ERP
⑩ Helium monitors 72-hour acceptance window and alerts seller's team if no response
```

Note on Accounts Payable Workflow: This diagram illustrates the Accounts Receivable (seller) workflow. When your organization is the buyer, Helium receives inbound invoice notifications via Taxly webhooks, initiates internal acceptance workflows for your AP team, and orchestrates your acceptance response back through Taxly to FIRS. The workflow is essentially reversed, with Helium managing your internal buyer-side processes rather than the seller-side submission process shown above.

1.2 The Post-November 2025 Reality

Current Compliance Landscape

The FIRS e-invoicing mandate is now in effect, requiring all businesses with annual turnover ≥₦5 billion to transmit invoices through the FIRS platform for pre-clearance before delivery to buyers. Businesses must establish compliant invoice workflows to mitigate regulatory risk and ensure uninterrupted operations.

Whether you're starting compliance now or enhancing existing processes, Helium streamlines your path forward. For businesses assessing their options, WML offers free interim solutions through **Vendra**—including bulk upload capabilities and a free vendor portal—allowing you to achieve basic compliance while planning your Helium implementation. This flexible approach ensures businesses can meet immediate regulatory requirements while building toward full automation.

FIRS Requirements Summary

The mandate encompasses two operational models:

Pre-Clearance Model (B2B/B2G Transactions):

- Helium generates Invoice Reference Number (IRN) and QR code
- Submit invoice to FIRS before sending to buyer
- FIRS validates invoice structure and business rules
- FIRS digitally signs invoice upon validation
- Transmit FIRS-signed invoice to buyer
- Track buyer acceptance within 72-hour window

Post-Audit Model (B2C Transactions):

- Issue invoice to consumer immediately
- Report invoice to FIRS within 24 hours
- Receive validation confirmation
- Maintain audit trail for FIRS review

Document Type Support Required:

- 381 (Standard Invoice)
- 380 (Commercial Invoice)
- 384 (Corrected Invoice)

- 385 (Self-Billed Invoice)
- 388 (Tax Invoice)
- 389 (Simplified Invoice)
- B2C Batch Reporting

The ERP Integration Challenge

Standard ERP systems were not designed for Nigeria's e-invoicing requirements. Direct ERP modification is complex, costly , time-intensive (6-12 months implementation), and creates permanent vendor dependencies. Additionally, modifying core ERP systems introduces:

- System instability risks during upgrades
- Increased testing overhead for every ERP patch
- Vendor support complications
- Difficulty migrating to new ERP systems

Helium provides an alternative approach: a middleware layer that bridges your existing ERP and FIRS compliance without touching your core systems. This architecture preserves ERP stability, maintains vendor support, and provides flexibility for future technology changes.

1.3 Versioned Rollout Strategy

Phased Delivery Model

Helium is delivered through a phased rollout model, ensuring customers achieve compliance while gaining enhanced automation and intelligence over time. This approach balances urgent regulatory needs with long-term operational excellence.

1.3.1 | Version 1.0 - Core Compliance (Available 8-12 Weeks from Contract Signing)

Version 1.0 delivers FIRS compliance with automated invoice submission, IRN and QR code generation, and ERP synchronization. This version supports all 7 FIRS document types and provides basic acceptance logging and notifications.

Core Capabilities:

Invoice Submission & Document Type Support:

- Automated transformation from ERP format to FIRS UBL 3.0 JSON
- Support for all 7 document types: 381, 380, 384, 385, 388, 389, B2C batch reporting
- Real-time validation before FIRS submission
- Automatic retry logic for transient failures
- Comprehensive error logging and alerting

IRN and QR Code Management:

- Helium generates Invoice Reference Numbers (IRNs) before submission to FIRS
- Helium generates QR codes containing IRN data
- Validates both IRN and invoice structure with FIRS through Taxly (APP)
- Stores IRN and QR code associations
- Maintains complete audit trail of invoice lifecycle

ERP Synchronization:

- Write-back of IRN, QR code, and acceptance status to ERP (requires ERP API capabilities)
- Configurable field mapping to ERP invoice tables

- Queue-based architecture ensures no data loss
- Retry mechanisms for failed write-back operations

Acceptance Logging (v1.0 Approach):

- Buyers can accept invoices through standard FIRS/APP channels (Querying APP portal or through email confirmation)
- Helium logs acceptance status when received via FIRS webhooks
- Email notifications sent to sellers with acceptance/rejection details
- Notifications include action buttons for quick response (no dashboard login required)
- Helium dashboard provides system health monitoring, audit logs, and compliance reporting - not daily workflow operations
- Manual follow-up required for non-responsive buyers

Basic Notifications:

- Email alerts to designated recipients upon invoice submission
- Status change notifications (accepted, rejected, error conditions)
- Optional storage to configured repository (shared drive, document management system, database, NAS, SAN)
- Configurable notification templates

Integration Support:

- API integration for real-time processing
- SFTP file-based integration for batch operations
- Hybrid deployment (both methods simultaneously)
- Detailed integration documentation provided

Operational Features:

- Web-based dashboard for system health monitoring and audit trails
- Search and filter capabilities for compliance audits
- Export functionality for regulatory reporting
- System analytics (submission rates, acceptance rates, error tracking)
- Polling capability: Helium can query FIRS via TIN lookup as fallback to webhooks for invoice status verification

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1.3.2 | Version 1.1 - Enhanced Workflow (Q1 2026)

Version 1.1 introduces intelligent automation that actively manages the invoice acceptance lifecycle. Rather than passively logging acceptance status, Helium initiates proactive workflows, sends automated reminders, and handles exceptions—eliminating manual follow-up by your teams.

Enhanced Capabilities:

Helium-Initiated Acceptance Workflow (Key Differentiator):

v1.0 enables invoice acceptance through standard FIRS channels. v1.1 adds intelligent automation—Helium actively manages the acceptance lifecycle, sends reminders, and handles exceptions so your team doesn't have to.

- **Automatic 72-Hour Countdown Tracking**: Helium begins monitoring acceptance window immediately upon successful invoice transmission to buyer
- **Proactive Reminder System**:
  - 24-hour reminder (day 1 after transmission)
  - 48-hour reminder (day 2 after transmission)
  - 68-hour final reminder (4 hours before deadline)
- **Multi-Channel Delivery**: Reminders delivered via email, ERP notifications, or configured repositories (file server, NAS, SAN, database)
- **Smart Escalation**: Automatic escalation to CRM if no response received by deadline
- **Seller Alert Management**: Your teams receive exception alerts only—not routine status updates—reducing notification fatigue
- **Configurable Workflows**: Customize reminder timing, escalation rules, and notification recipients per buyer or buyer segment

Dual-Channel Notification Architecture:

Helium v1.1 implements redundant notification channels to ensure buyers receive invoice notifications even if primary systems fail.

- **Primary Channel - API Notifications**: Direct API calls to buyer's ERP or invoice management system (where available)
- **Fallback Channel - SMTP Email**: Automatic email notification if API unavailable or fails
- **Channel Priority Logic**: Configurable preference order (API first, email backup, or email only)
- **Retry Mechanisms**: Exponential backoff for failed deliveries
- **Delivery Confirmation**: Tracking of successful delivery via API acknowledgment or email read receipts (where supported)
- **Notification Content**: Structured data for API consumers, rich HTML for email recipients, includes invoice summary and acceptance link

Portobello: Buyer Onboarding & Exception Management:

The Portobello workflow automates the complex process of onboarding unregistered buyers to the FIRS platform. Portobello triggers when FIRS validation detects an unregistered buyer, temporarily stalling invoice submission until the buyer completes FIRS Merchant Buyer Solution (MBS) registration. This process flow utilizes system and human swim-lanes to clearly delineate automated processes from manual interventions, ensuring transparency and accountability.

Automatic Detection:

- Helium identifies unregistered buyers through FIRS validation errors
- Flags invoices requiring special handling
- Creates exception queue for processing
- Stalls invoice submission until buyer registers on FIRS MBS portal

Guided Onboarding Workflow:

- **System Swim-Lane**: Automated email communication to buyer explaining registration requirements
- **Human Swim-Lane**: Manual follow-up by seller's team if buyer doesn't respond within 48 hours
- Step-by-step registration guidance with FIRS portal links
- Progress tracking through onboarding stages
- Notification triggers at key milestones (registration started, documents submitted, registration completed)

CRM Integration:

- Automatic ticket creation in supported CRM systems (Zendesk, Freshdesk, ServiceNow, Salesforce, Microsoft Dynamics CRM)
- Tickets include buyer information, invoice details, and onboarding status
- Automatic updates as buyer progresses through registration
- Escalation rules for stalled onboarding

Onboarding Webhooks:

- Helium receives webhook notifications from FIRS when buyer completes registration
- Automatic transition from "pending registration" to "active" status
- Triggers resumption of invoice submission workflow

Seamless Transition:

- Once buyer completes FIRS registration, Helium automatically resumes standard acceptance workflow
- Historical invoices can be resubmitted for acceptance
- No data loss or manual reconciliation required

1.3.3 | Version 2.0 - AI & Advanced Features (Q2 2026)

Version 2.0 leverages AI to provide predictive insights, self-service capabilities, and advanced exception handling. This release transforms Helium from a compliance tool into a strategic platform for invoice operations optimization.

Advanced Capabilities:

AI-Powered Data Mapping:

- Intelligent field mapping suggestions based on ERP schema analysis
- Pattern recognition from historical invoice data
- Learning system improves recommendations over time

Predictive Compliance Analytics:

- Early warning system for potential compliance issues
- Buyer behavior pattern analysis
- Risk scoring for invoices based on historical patterns
- Proactive recommendations to improve acceptance rates

Advanced Exception Handling:

- AI-assisted resolution recommendations for rejection reasons
- Automatic categorization of rejection types
- Suggested corrective actions

Self-Service Buyer Portal:

- White-labeled portal for buyers to interact with invoices
- View details, download copies, accept/reject with reason codes
- Invoice history and dispute management
- Mobile-responsive with SSO integration

Enhanced Reporting & Dashboards:

- Real-time analytics and custom report builder
- Executive dashboards with KPIs (submission rates, acceptance times, rejection trends)
- Scheduled delivery and export capabilities
- API access for BI tool integration

1.4 Integration Options Overview

Integration Methods

Helium supports two primary integration methods, allowing organizations to choose the approach that best fits their existing infrastructure and operational requirements. Many customers deploy both methods simultaneously—API for real-time priority invoices and SFTP for batch processing of routine transactions.

1.4.1 Method 1: API Integration (Real-Time)

Optimal For:

- Modern ERPs with native automation capabilities (Dynamics 365, SAP S/4HANA, NetSuite, Oracle Cloud)
- Custom-built systems with REST API support
- Organizations requiring sub-second invoice processing
- High-volume environments where real-time status updates are critical

Architecture:

Your ERP's native automation layer (Power Automate workflows, SAP background jobs, Oracle Scheduled Processes, custom triggers) posts invoice data to Helium's REST API. Helium generates the IRN and QR code, processes the invoice, submits to FIRS via Taxly, receives digital signature, and writes back to your ERP—all within seconds.

Technical Requirements:

- **ERP Capabilities**: Outbound HTTP/HTTPS request capability via webhooks, scheduled jobs, or custom code
- **Network Connectivity**: HTTPS access to Helium (deployed in your infrastructure or WML cloud)
- **Data Format**: JSON request/response bodies (XML support available)
- **Authentication**: OAuth 2.0 or API key-based authentication
- **TLS**: Transport Layer Security 1.3 for all communications

Integration Flow:

```
1. ERP detects invoice ready for submission (trigger: invoice posted, workflow approved)
2. ERP automation layer calls Helium API endpoint: POST /api/v1/invoices
3. Helium generates IRN and QR code
4. Helium validates payload, transforms to UBL 3.0, submits to FIRS via Taxly
5. Helium returns response to ERP: { "irn": "...", "qrCode": "...", "status": "submitted" }
6. ERP automation updates invoice record with IRN and QR code
7. Helium sends webhook to ERP when acceptance status changes
```

Performance Characteristics:

- **Latency**: Real-time processing (typically under 1 second per invoice)
- **Reliability**: Automatic retry for transient failures
- **Scalability**: Horizontal scaling for peak periods

Infrastructure Costs:

- **ERP Integration Layer**: Free (if ERP has native API/automation capabilities)
- **Network**: Standard HTTPS traffic (minimal bandwidth)
- **Helium Deployment**: Included in subscription

Implementation Complexity:

- **Low**: ERPs with visual workflow builders (Power Automate, ServiceNow Flow Designer)
- **Medium**: ERPs requiring custom code but with REST client libraries (SAP ABAP, Oracle PL/SQL)
- **Timeline**: 2-4 weeks typical for API integration setup

1.4.2 Method 2: SFTP File-Based Integration (Batch)

Optimal For:

- Legacy ERPs without API capabilities (Quantum, Sage X3, older SAP R/3, Syspro)
- Organizations with established file-based workflows
- Batch processing environments (nightly invoice processing)
- Air-gapped or highly restricted network environments
- Systems where API development is cost-prohibitive

Architecture:

Your ERP exports invoice data to files (CSV, XML, or JSON format). A scheduler application (Oracle Scheduler, SQL Server Agent, Windows Task Scheduler, cron) automatically uploads files to Helium's SFTP server at configured intervals. Helium processes each invoice immediately upon file detection, generates IRN and QR code, submits to FIRS via Taxly, receives digital signature, and generates result files for ERP import.

Technical Requirements:

- **ERP Capabilities**: File export functionality (to CSV, XML, or JSON)
- **Scheduler**: Automated job scheduler (Oracle Scheduler, SQL Server Agent, cron, or manual)
- **SFTP Client**: Standard SFTP client for file transfer (built into most operating systems)
- **File Format**: Structured data files following Helium's schema (templates provided)
- **SFTP Server**: Customer-managed or WML-hosted SFTP server

Integration Flow:

```
1. ERP exports invoice(s) to file: /export/invoices_20251101_120000.csv
2. Scheduler uploads file to Helium SFTP: sftp://helium-sftp/inbound/
3. Helium detects new file, validates format, processes each invoice immediately
4. Helium generates IRN and QR code for each invoice
5. Helium submits to FIRS via Taxly, receives digital signatures
6. Helium generates result file: /outbound/results_20251101_120000.csv
7. Scheduler downloads result file to ERP import directory
8. ERP import process updates invoice records with IRN, QR codes, and status
```

Performance Characteristics:

- **Latency**: 15-30 minutes per batch (configurable frequency)
- **Batch Frequency**: Configurable (every 15 minutes, hourly, daily, or on-demand)
- **Processing**: Each invoice submitted immediately upon file detection

Infrastructure Costs:

- **SFTP Server**:
  - Customer-hosted: Free (use existing infrastructure)
  - WML-hosted: minimum ₦450,000/year (managed service with 99.9% uptime)
- **Scheduler**: Free (use existing ERP scheduler or OS-level scheduler)
- **Network**: Standard SFTP traffic (minimal bandwidth)

Implementation Complexity:

- **Medium**: Requires file format mapping, scheduler configuration
- **Timeline**: 3-6 weeks typical for SFTP integration setup
- **Skill Requirements**: Basic knowledge of file formats, SFTP, and scheduler configuration

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1.4.3 Hybrid Approach

Use Case:

Many organizations deploy both integration methods simultaneously to balance real-time requirements with batch efficiency:

Example Scenario - Manufacturing Company:

- **API Integration**: Customer invoices requiring immediate IRN for shipment (time-sensitive, high-priority)
- **SFTP Integration**: Supplier invoices processed in nightly batch (routine, non-urgent)
- **Result**: Optimized processing with both methods writing back to same ERP invoice tables

Benefits of Hybrid:

- **Flexibility**: Choose integration method per invoice type, customer segment, or urgency
- **Risk Mitigation**: If API unavailable, fallback to SFTP for business continuity
- **Cost Optimization**: Use free API for most invoices, reserve SFTP for legacy systems requiring file-based approach
- **Migration Path**: Start with SFTP during implementation, transition high-volume invoices to API gradually

Configuration:

Helium's architecture supports simultaneous operation of both integration methods with unified configuration, monitoring, and reporting. No additional licensing or infrastructure required.

1.5 Why Helium? Core Differentiators

Strategic Value Propositions

Helium's architecture is purpose-built to solve the unique challenges of Nigerian e-invoicing compliance without disrupting existing business systems. The following differentiators address the primary concerns of IT leadership when evaluating compliance solutions.

1.5.1 ERP Abstraction - Zero Core System Modification

The Primary Value Proposition for IT Leaders:

Helium operates as an external orchestration layer, requiring no changes to your ERP's core logic, database schema, stored procedures, or business rules. Your ERP remains stable, supportable, and upgrade-ready.

What This Means in Practice:

- **No Custom Code in ERP**: Helium does not require custom tables, triggers, or stored procedures in your ERP database
- **No ERP Vendor Involvement**: Implementation proceeds without engaging your ERP vendor for core system changes
- **Vendor Support Preserved**: ERP vendor support agreements remain valid—no "unsupported customizations" that void warranties
- **Upgrade Path Protected**: Apply ERP patches, service packs, and version upgrades without regression testing Helium integration
- **Audit Trail Independence**: Helium maintains its own audit logs independent of ERP, simplifying compliance reviews

Technical Implementation:

Helium integrates through standard interfaces only:

- **API Method**: ERP's native automation layer (Power Automate, SAP jobs) calls Helium REST endpoints
- **SFTP Method**: ERP exports files using standard export tools, no custom development required
- **Write-Back**: Helium updates ERP through APIs (where available) or import files—never direct database writes

Business Continuity Benefit:

If you migrate to a new ERP system in the future, Helium adapts to the new integration interfaces without replacing the entire compliance infrastructure. Your compliance layer remains constant even as underlying systems change.

1.5.2 Universal ERP Compatibility

Platform Agnostic by Design:

Unlike vendor-specific e-invoicing modules that lock you into one ERP ecosystem (and one vendor's pricing model), Helium works with any system through standard integration patterns.

Supported ERP Systems (Non-Exhaustive List):

- **Microsoft Dynamics**: 365 Business Central, Finance & Operations, NAV, GP
- **SAP**: S/4HANA, Business One, older R/3 versions
- **Oracle**: Cloud ERP, E-Business Suite, JD Edwards, PeopleSoft
- **Sage**: X3, Intacct, 300, 50cloud
- **Infor**: CloudSuite, LN, M3
- **Custom-Built Systems**: Any system with file export or API capabilities

Integration Flexibility:

- **Modern ERPs**: Use native automation (Power Automate, Oracle Integration Cloud, SAP Cloud Platform) to call Helium API
- **Legacy ERPs**: Export files, use scheduler to upload via SFTP
- **Hybrid Environments**: Different integration methods per ERP instance in multi-system organizations

Future-Proofing:

Organizations planning ERP migrations benefit from Helium's universal compatibility. Examples:

- **Current State**: Sage X3 with SFTP integration
- **Future State**: Migrate to Dynamics 365 with API integration
- **Helium Role**: Remains constant compliance layer—only integration method changes, core functionality preserved

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1.5.3 Flexible Deployment Options

Infrastructure Control:

Helium's containerized architecture supports deployment in your infrastructure or WML's managed cloud—you choose based on data sovereignty, security, and operational requirements.

Deployment Models:

Customer-Hosted (On-Premise or Private Cloud):

- Deploy Helium in your data center or private cloud (Azure, AWS, Google Cloud)
- Complete control over infrastructure, security policies, network configuration
- Data never leaves your environment—meets strictest data sovereignty requirements
- You manage: Infrastructure, OS patches, backup/recovery, monitoring
- WML provides: Container images, deployment scripts, configuration guidance, application support

WML-Hosted (Managed Cloud):

- Helium runs in WML's Azure West Africa infrastructure
- WML manages: Infrastructure, security patches, backup/recovery, monitoring, scaling
- You retain: Data ownership, access controls, configuration management
- Reduced operational overhead—focus on business processes, not infrastructure management
- 99.9% uptime SLA with 24/7 monitoring

Hybrid Deployment (Advanced):

- Core Helium services in your infrastructure
- Optional WML-managed components (SFTP server, monitoring dashboards)
- Balance control and convenience

Decision Criteria:

Consideration	Customer-Hosted	WML-Hosted
Data Sovereignty	Full control	WML manages (Nigeria-based)
Operational Overhead	IT team manages	WML manages
Infrastructure Costs	Customer bears	Included in subscription
Customization Flexibility	Full control	Standard deployment
Best For	Large enterprises, regulated industries	Mid-sized businesses, fast deployment

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1.5.4 Phased Value Delivery

Compliance Implementation with Continuous Enhancement:

Rather than waiting 6-12 months for a complete solution, Helium delivers value in stages:

- **Weeks 8-12 (v1.0)**: Begin with core compliance implementation (IRN generation, submission, ERP synchronization)

- **Q1 2026 (v1.1)**: Gain workflow automation (acceptance tracking, notifications, Portobello)
- **Q2 2026 (v2.0)**: Unlock AI-powered intelligence (predictive analytics, self-service portal)

Included in Enterprise Subscription:

Helium Enterprise customers receive all versions (v1.0 → v2.0) as they release—no additional licensing fees, no upgrade projects, no re-implementation. As new capabilities become available, WML deploys updates to your Helium instance with zero downtime.

Business Value Progression:

- **Month 1-3**: Compliance achieved, manual processes eliminated
- **Month 4-6**: Workflow automation significantly reduces finance team workload
- **Month 7-12**: AI insights optimize invoice acceptance rates, identify process improvements

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1.5.5 Professional Services & Consultation Included

Implementation Success Built-In:

Helium is not just software licensing—WML's implementation includes comprehensive consulting services to ensure successful deployment and ongoing compliance management.

Included Consultation Services:

Pre-Implementation Phase:

- **Readiness Assessment (4 hours)**: Evaluate current ERP environment, data quality, integration capabilities, and compliance gaps
- **Compliance Oversight Planning (2 hours)**: Define compliance monitoring processes, audit trail requirements, and regulatory reporting workflows
- **Data Mapping Workshop (4 hours)**: Deep dive on FIRS requirements and ERP data structure analysis
- **Schema Alignment Review (2 hours)**: Ensure ERP invoice data structure maps correctly to FIRS UBL 3.0 JSON schema

Implementation Phase:

- **ERP Integration Consulting (4 hours)**: Technical architecture review, API/SFTP specification walkthrough, and integration pattern selection
- **Project Consulting**: Weekly 30-minute calls during 8-12 week implementation phase for Q&A, guidance, and progress tracking
- **Risk and Issue Tracking**: Dedicated project management support for identifying, escalating, and resolving implementation risks

Testing & Go-Live Phase:

- **UAT & Certification (4 hours)**: FIRS Sandbox testing, validation, and production readiness certification

- **Scheduler Configuration**: Oracle Scheduler, SQL Server Agent, or cron setup assistance (SFTP method)
- **Go-Live Support**: Intensive first-week monitoring and issue resolution

Scope Boundaries:

WML provides Helium configuration and integration guidance. WML does not debug ERP internal code, develop custom integrations outside standard patterns, or provide on-site support visits (unless contracted separately). Additional consulting hours available at published rates.

1.5.6 Built on Proven Technology

Production-Grade Reliability:

Helium's core architecture is based on technology proven in Nigerian regulatory environments:

ATRS Foundation:

The Accreditation and Tax Remittance System (ATRS) demonstrates WML's capability to build production-grade compliance platforms. While Helium is a distinct product designed specifically for e-invoicing, it leverages architectural patterns, security frameworks, and operational practices validated through ATRS deployment.

Key Similarities:

- Regulatory data transformation and validation
- High-volume transaction processing
- Audit logging and compliance reporting
- Integration with government platforms

Technical Maturity:

Helium is not a startup proof-of-concept—it's built by a team with:

- 5+ years experience in Nigerian regulatory technology
- Production systems processing millions of transactions monthly
- Deep understanding of FIRS requirements and platform behavior
- Established relationships with regulatory authorities

1.6 Document Scope & Engagement Process

Preview Documentation Package

WestMetro provides three documents for initial Helium evaluation:

1.6.1 Module 1: Executive Overview & Phased Roadmap (This Document)

Purpose:
Provides high-level understanding of Helium's capabilities, versioned roadmap, integration options, and core differentiators.

Intended Audience:
CIOs, IT Directors, Solutions Architects, and business stakeholders evaluating e-invoicing compliance solutions.

Content:
Executive summary, phased delivery model (v1.0, v1.1, v2.0), integration methods (API and SFTP), value propositions, and engagement process.

1.6.2 Module 2: Security & Compliance Architecture (Separate Document)

Purpose:
Detailed security architecture, encryption standards, network security, audit logging, SIEM integration, compliance certifications, and vulnerability management.

Intended Audience:
Chief Information Security Officers (CISOs), security teams, compliance officers, and internal audit teams.

Content:

- Security architecture overview
- Data encryption (at rest and in transit)
- Network security (firewalls, VPN, DMZ architecture)

- Authentication and access control (MFA, RBAC, OAuth 2.0)
- Audit logging and SIEM integration (comprehensive audit trails, log retention, tamper-proof logging)
- Compliance certifications (SOC 2, ISO 27001 roadmap, NDPR compliance)
- Vulnerability management (penetration testing, CVE monitoring, incident response)
- Data residency and sovereignty options

Note: Module 2 is provided as a separate document to facilitate independent security review workflows.

1.6.3 Complete Technical Specification (Provided Upon Formal Engagement)

Upon engagement, WestMetro provides the complete Helium Technical Specification v3.1 (Modules 1-11), which includes:

```
Module 1:  Executive Overview & Phased Roadmap (preview provided)
Module 2:  Security & Compliance Architecture (preview provided)
Module 3:  System Architecture & Components (queue architecture, high-availability options,
           deployment topologies)
Module 4:  Data Flow & Invoice Lifecycle
Module 5:  API Integration Method
Module 6:  SFTP File-Based Integration
Module 7:  Hybrid Dual-Channel Approach
Module 8:  Core Workflows & Operations
Module 9:  Observability, Monitoring & Operations
Module 10: Implementation Guide & Rollout
Module 11: Appendices (ERP Compatibility Matrix, API Reference, File Format Specifications,
           Security Questionnaire)
```

Module 3-11 Content Summary:

- **Modules 3-4**: Detailed system architecture, component specifications, data flow diagrams, state machines
- **Modules 5-7**: Step-by-step integration implementation guides with code examples, network diagrams, configuration templates
- **Module 8**: Workflow specifications, Portobello process flows, acceptance management, exception handling procedures
- **Module 9**: Monitoring stack configuration, observability tools, SLA definitions, incident response procedures
- **Module 10**: Implementation timeline, resource requirements, testing plans, go-live checklists, pricing and edition comparison
- **Module 11**: Technical reference materials, API endpoint documentation, ERP-specific integration notes

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1.6.4 Evaluation Process

Step 1: Business Review
Use this document (Module 1) to assess Helium's fit with your compliance strategy, IT architecture, and operational requirements.

Key Questions to Answer:

- Does Helium's phased delivery model align with your compliance timeline?
- Which integration method (API, SFTP, or hybrid) fits your ERP environment?
- Does the zero-modification approach preserve your ERP stability requirements?
- Is the deployment model (customer-hosted vs. WML-hosted) appropriate for your data sovereignty needs?

Step 2: Security Review
Share Module 2 (separate document) with your CISO, security team, and compliance officers for detailed security evaluation.

Key Questions to Answer:

- Does Helium's security architecture meet your information security policies?
- Are encryption standards, audit logging, and SIEM integration sufficient?
- Does the deployment model support your data residency requirements?
- Are planned certifications (SOC 2, ISO 27001) aligned with your compliance needs?

Step 3: Technical Deep-Dive
Upon positive initial assessment and contractual agreements, request complete specification (Modules 3-11) to begin detailed integration planning.

Activities Include:

- Review system architecture and component specifications (Module 3)
- Analyze data flow and invoice lifecycle (Module 4)
- Select integration method and review implementation guide (Modules 5-7)
- Assess operational requirements and monitoring capabilities (Module 9)
- Plan implementation timeline and resource allocation (Module 10)

Step 4: Formal Engagement
Sign NDA and engagement agreement to receive Modules 3-11 and begin implementation scoping.

Engagement Deliverables:

- Complete technical specification (Modules 1-11)
- Access to WML technical team for Q&A
- Data mapping workshop scheduling
- Implementation project plan development

- Sandbox environment access for testing

1.6.5 Next Steps

To request complete documentation or discuss Helium implementation:

WestMetro Limited
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Typical Engagement Timeline:

```
Week 1-2:   Initial consultation, security review, NDA execution
Week 3:     Receive complete specification (Modules 3-11)
Week 4:     Data mapping workshop, integration planning
Week 5-8:   Development and configuration (customer-side)
Week 9-10:  UAT in FIRS Sandbox environment
Week 11:    Production deployment preparation
Week 12:    Go-live with intensive support
```

Pre-Engagement Preparation:

To accelerate evaluation and implementation, consider preparing:

- Sample invoice data exports from your ERP (anonymized)
- ERP system version and integration capability documentation
- Data sovereignty and security policy requirements
- Expected invoice volume projections
- List of buyer TINs requiring onboarding (if known)

End of Module 1

Document Statistics:

- Module: 1 of 11
- Version: 3.1
- Pages: ~25
- Last Updated: November 2025

Related Documents:

- Module 2: Security & Compliance Architecture (separate document)
- Complete Technical Specification: Modules 3-11 (provided upon engagement)

For Questions or Complete Documentation:
Contact: info@westmetrong.com

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